

5.8.9

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Question

The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle

Solution

Let the original length and breadth of the rectangle be l and b , respectively.

From the problem:

$$(l - 5)(b + 3) = lb - 9 \quad (1)$$

$$\Rightarrow 3l - 5b = 6 \quad (2)$$

$$(l + 3)(b + 2) = lb + 67 \quad (3)$$

$$\Rightarrow 2l + 3b = 61 \quad (4)$$

Equations (2) and (4) form the system:

$$\begin{pmatrix} 3 & -5 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} l \\ b \end{pmatrix} = \begin{pmatrix} 6 \\ 61 \end{pmatrix} \quad (5)$$

Solution

With the augmented matrix followed by row reduction:

$$\left(\begin{array}{cc|c} 3 & -5 & 6 \\ 2 & 3 & 61 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - \frac{2}{3}R_1} \left(\begin{array}{cc|c} 3 & -5 & 6 \\ 0 & \frac{19}{3} & 57 \end{array} \right) \quad (6)$$

$$\xrightarrow{R_2 \leftarrow \frac{3}{19}R_2} \left(\begin{array}{cc|c} 3 & -5 & 6 \\ 0 & 1 & \frac{171}{19} \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + 5R_2} \left(\begin{array}{cc|c} 3 & 0 & \frac{969}{19} \\ 0 & 1 & \frac{171}{19} \end{array} \right) \quad (7)$$

$$\xrightarrow{R_1 \leftarrow \frac{1}{3}R_1} \left(\begin{array}{cc|c} 1 & 0 & \frac{323}{19} \\ 0 & 1 & \frac{171}{19} \end{array} \right) \quad (8)$$

Solution

$$\therefore \begin{pmatrix} l \\ b \end{pmatrix} = \begin{pmatrix} \frac{323}{19} \\ \frac{171}{19} \end{pmatrix} = \begin{pmatrix} 17 \\ 9 \end{pmatrix} \quad (9)$$

The dimensions of the rectangle are:

Length = 17 units, Breadth = 9 units

Code (C)

Part 1

```
#include <stdio.h>

void solve(int* l, int* b) {
    int i, j;
    for (i = 0; i <= 100; i++) {
        for (j = 0; j <= 100; j++) {
            if ((3 * i - 5 * j == 6) && (2 * i + 3 * j == 61)) {
                printf("Found solution: l = %d, b = %d\n", i, j);
                *l = i;
                *b = j;
                return;
            }
        }
    }
    printf("No solution found.\n");
    *l = -1;
    *b = -1;
}
```

Part 1

```
import ctypes

# Load DLL (adjust filename/path as needed)
lib = ctypes.CDLL('./cod10.dll')

# Declare function argument and return types
lib.solve.argtypes = [ctypes.POINTER(ctypes.c_int), ctypes.
    POINTER(ctypes.c_int)]
lib.solve.restype = None

# Prepare ctypes int variables to hold results
l = ctypes.c_int()
b = ctypes.c_int()
```

Part 2

```
# Call the solve function from the DLL
lib.solve(ctypes.byref(l), ctypes.byref(b))

length = l.value
breadth = b.value

if length == -1 or breadth == -1:
    print("No valid solution found.")
else:
    print(f"Length = {length}, Breadth = {breadth}")
```